Project Executable Files

Date	5 July 2026
Team ID	SWTID-2026-4860
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	yes
9	Test files and test results	NA
10	Demo video or walkthrough (if required)	NA

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Step 2: File / Folder Structure

```
EDUGENIE/
│
├── static/
│   └── style.css              - Application layout stylesheets & CSS loading animations
├── templates/
│   └── index.html             - Core user interface (HTML5, Vanilla JS, & Jinja2 Templates)
├── Dockerfile                 - Container configuration script for Hugging Face Spaces deployment
├── explanation_module.py      - Concept breakdown logic using Google Gemini 2.5 Flash
├── learning_path.py           - Personalized academic timeline generation module
├── main.py                    - Main FastAPI application server entry point & routing engine
├── qna.py                     - Context-based Question Answering execution pipeline
├── quiz_module.py             - Interactive test generation script (with local fallback logic)
├── requirements.txt           - Core Python dependencies and system requirements
├── README.md                  - Detailed local setup and user workflow instructions
└── summary_module.py          - Text compression and summarization pipeline module
```

Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	https://harsh0819-edugenie-assistant.hf.space
Login Credentials (Demo)	No authentication required
Platform / Hosting Provider	Hugging Face Spaces
Repository Link	https://github.com/Harsh-Saini-0819/EduGenie-Google-Gemini-Powered-Learning-Assistant.git
Demo Video Link	NA

Step 4: Run Instructions

Pre-requisites:

Python: Version 3.10+
Credentials: Google Gemini API Key
Network: Active internet connection

Step 1: Clone the repository using
git clone "https://github.com/Harsh-Saini-0819/EduGenie-Google-Gemini-Powered-Learning-Assistant.git"

Step 2: Open terminal and navigate into the cloned project directory.

Step 3: Create and activate a Python virtual environment.

Windows: python -m venv venv && venv\Scripts\activate
Mac/Linux: python -m venv venv && source venv/bin/activate

Step 4: Install requirements with pip install -r requirements.txt.

Step 5: Run configuration command

Windows CMD: set GEMINI_API_KEY=your_actual_api_key_here
Windows PowerShell: $env:GEMINI_API_KEY="your_actual_api_key_here"
Mac/Linux: export GEMINI_API_KEY="your_actual_api_key_here"

Step 6: Start the backend server using
uvicorn main:app --reload.

Step 7: Open your browser and go to http://127.0.0.1:8000.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	**HTTP 503 Service Unavailable** Google API endpoints can occasionally reject requests during high global traffic peak hours if using a Free Tier API key	**Workaround:** Switch the model parameter in the Python scripts from gemini-2.5-flash to the highly available gemini-2.5-flash-lite, or wait 2 minutes and click "Run Engine" again.
2	**Initial Local Model Load Delay** Executing the quiz fallback system locally using CPU resources can experience an initialization latency while loading the LaMini-Flan-T5 token weights into memory.	**Status:** Resolved by caching weights locally after the first initialization. Users are advised to rely primarily on the connected cloud API pipeline for instant interactive generation.
3	**Missing API Key Authorization Fault** Opening a new isolated terminal thread resets current session data, causing the application modules to lose access to the system key.	**Workaround:** Re-execute environment injection command explicitly in the new terminal, prior to launching Uvicorn.